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CE F231

Fluid Mechanics

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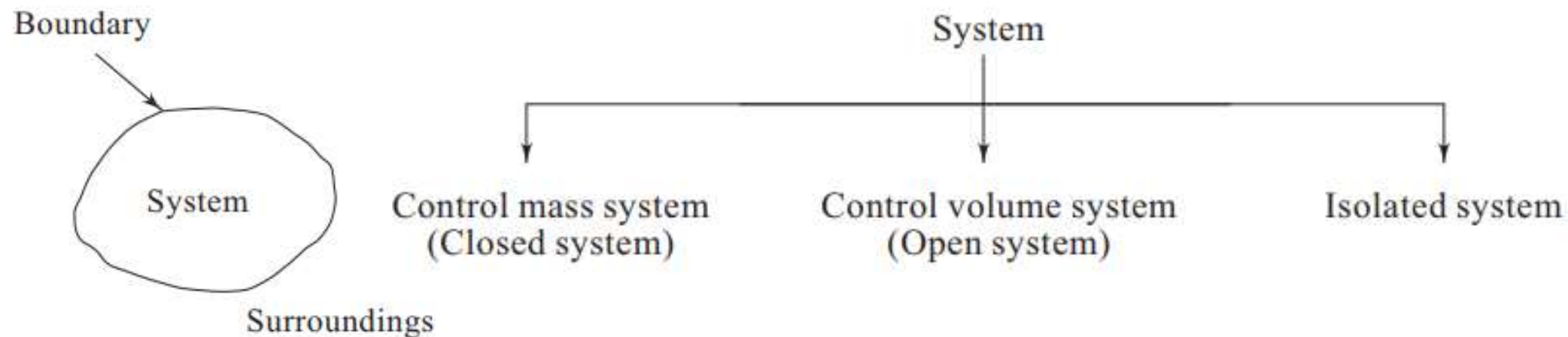
Lecture 18: Fluid Kinematics

October 18, 2022

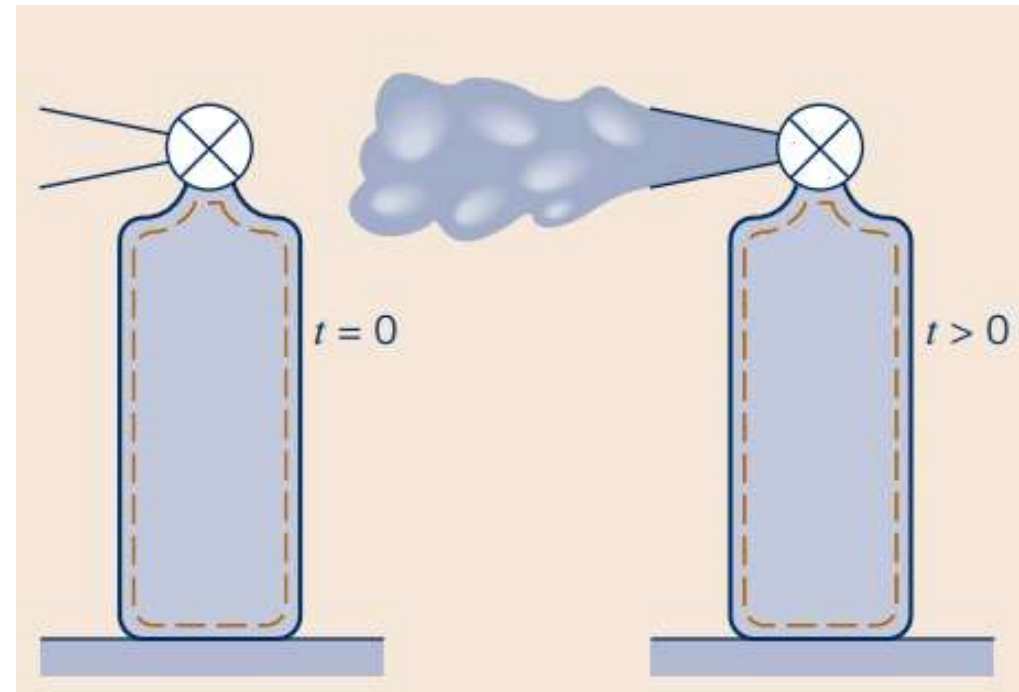
Recap: System

➤ System

- A system is defined as a quantity of matter in space on which attention is paid in the analysis of problem
- Everything outside system is called surroundings

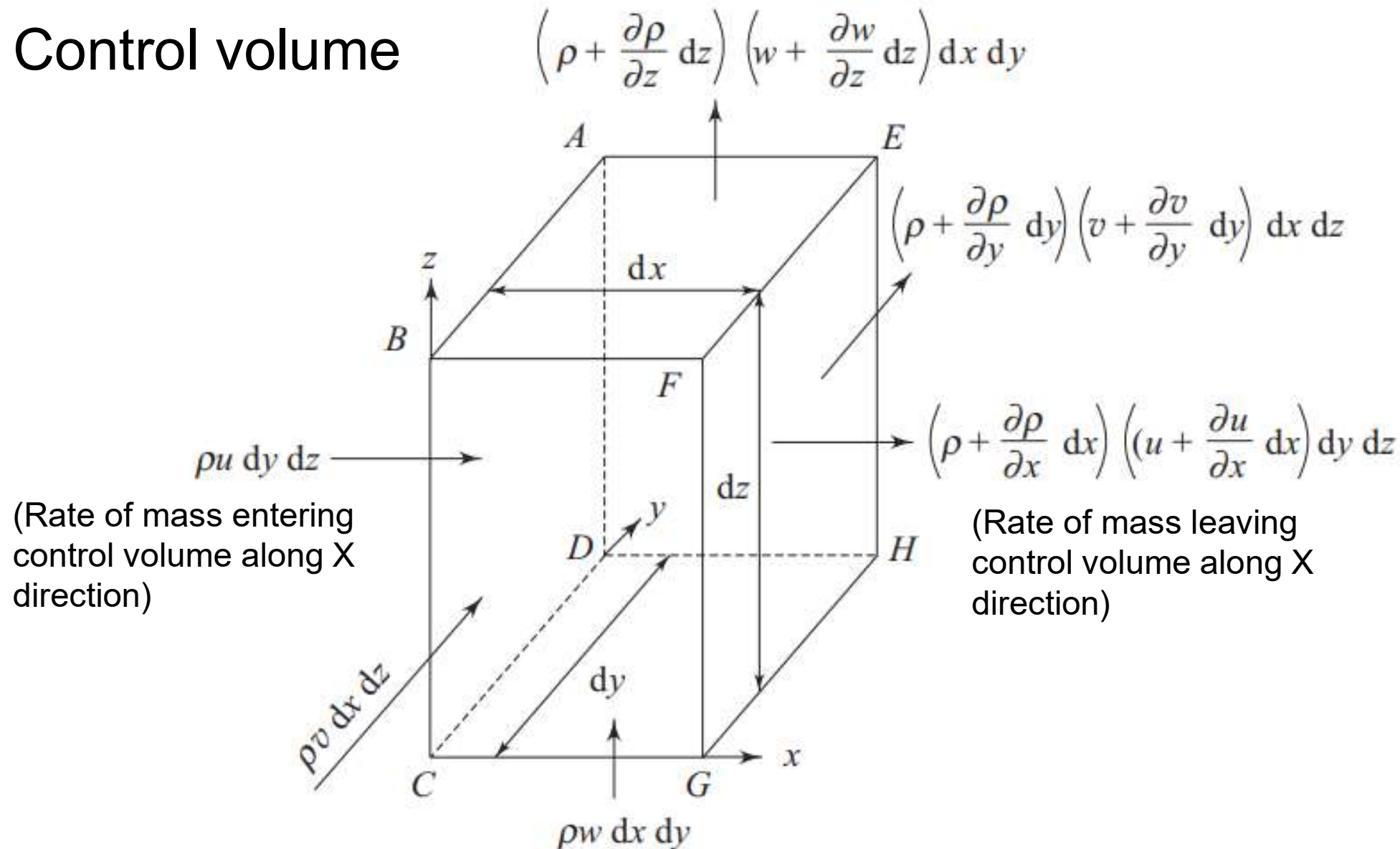


Recap: System and Control Volume



Recap: Conservation of Mass – The Continuity Equation

➤ Control volume



Recap: Conservation of Mass – The Continuity Equation

Rate of accumulation of mass in the control volume +
Net rate of mass efflux from the control volume = 0

The rate of accumulation of mass within the control volume is $\frac{\partial}{\partial t} (\rho \, dV) =$

$\frac{\partial \rho}{\partial t} \, dV$ (by the definition of control volume, dV is invariant with time).

Along X - direction

$$\frac{\partial}{\partial x} (\rho u) \, dV$$

Along Y - direction

$$\frac{\partial}{\partial y} (\rho v) \, dV$$

Along Z - direction

$$\frac{\partial}{\partial z} (\rho w) \, dV$$

$$\left\{ \frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x} (\rho u) + \frac{\partial}{\partial y} (\rho v) + \frac{\partial}{\partial z} (\rho w) \right\} dV = 0$$

Example

A conical nozzle 0.5 m long converging from 0.4 m to 0.2 m diameter linearly is subjected to an outlet flow varying linearly from 0 to 1 m³/s in 10 seconds. Determine the total acceleration at the mid-length of the nozzle at the mid-time of variation. Assume uniform flow over each cross-section.

Find discharge at outlet as function of t

Find diameter of any section as function of x

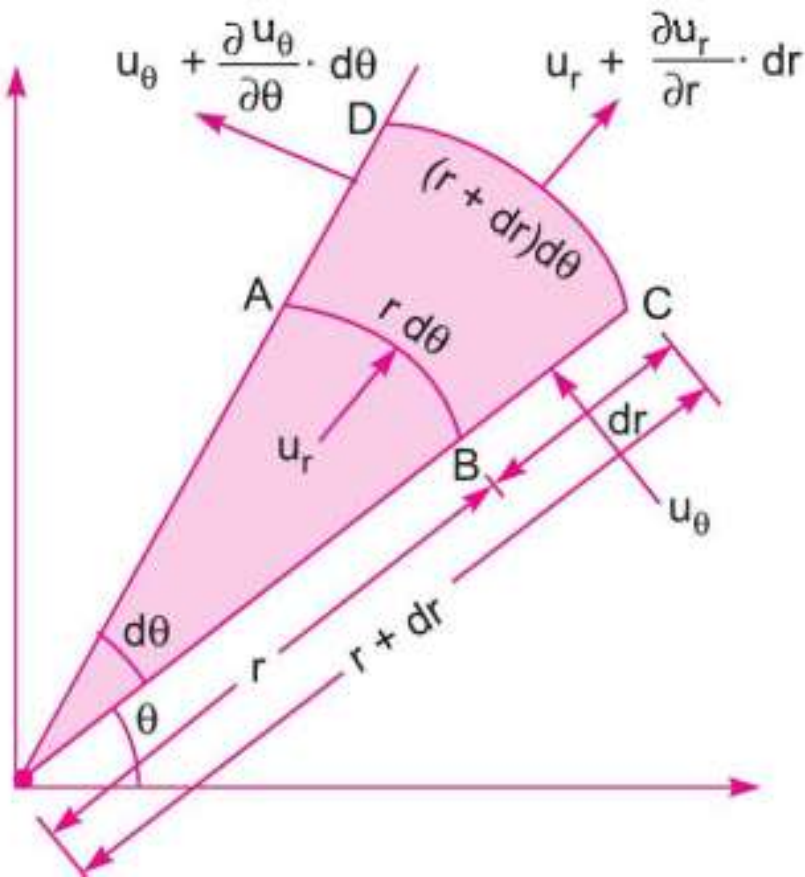
Find discharge at any section as function of t

Find velocity at any section as function of x and t

Find total acceleration

Reading Assignment

Continuity Equation in Cylindrical Polar Coordinates (2D flow)



Rate of accumulation of mass in the control volume +
Net rate of mass efflux from the control volume = 0

$$\frac{\partial \rho}{\partial t} + \rho \left[\frac{u_r}{r} + \frac{\partial u_r}{\partial r} \right] + \rho \frac{\partial u_\theta}{\partial \theta} \cdot \frac{1}{r} = 0$$

$$u_r + r \frac{\partial u_r}{\partial r} + \frac{\partial u_\theta}{\partial \theta} = 0$$

Stream Function (Ψ - psi)

- Scalar function; defined for 2D flow
- Partial derivative of Ψ with respect to any direction (x or y) gives velocity component at right angle to that direction
- Satisfies continuity eqn. for 2D incompressible flow

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0.$$

$$\frac{\partial \Psi}{\partial x} = v$$

$$\frac{\partial \Psi}{\partial y} = -u$$

OR

$$v = -\frac{\partial \Psi}{\partial x}$$

$$u = \frac{\partial \Psi}{\partial y}$$

Stream Function (Ψ - psi)

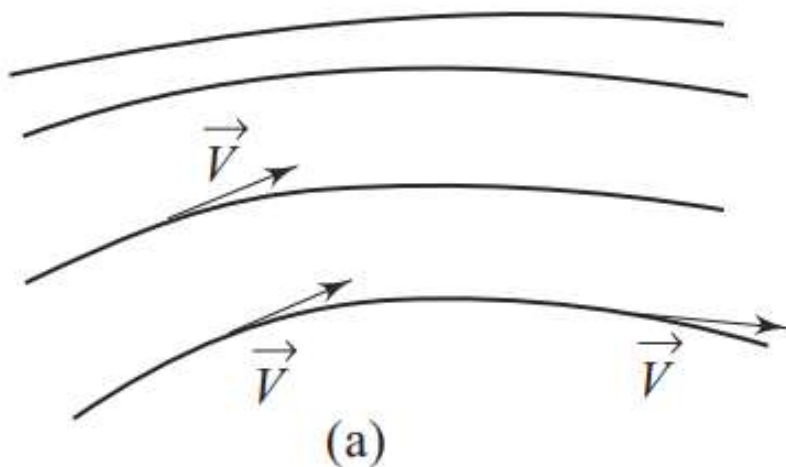
$$d\psi = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy = -v dx + u dy$$

$\psi = \text{Constant}$
 $d\psi = 0$ (Equi-stream function line)

$$v = -\frac{\partial \psi}{\partial x}$$

$$u = \frac{\partial \psi}{\partial y}$$

Streamlines

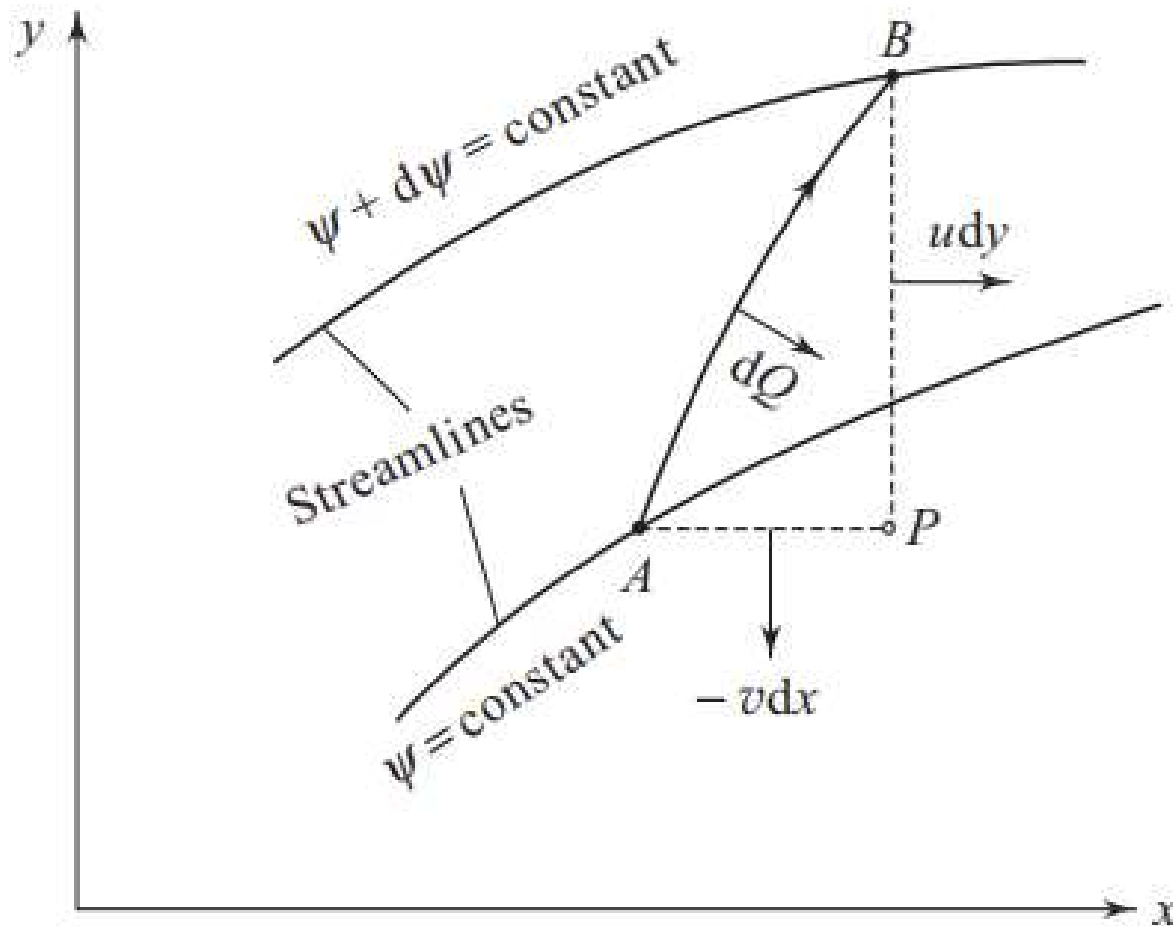


$$\vec{V} \times d\vec{S} = 0$$

$$\frac{dx}{u} = \frac{dy}{v}$$

Value of stream function does not change along a streamline

Physical Significance of Stream Function



$$dQ = u \, dy - v \, dx$$

$$u = \frac{\partial \psi}{\partial y} \quad v = -\frac{\partial \psi}{\partial x}$$

$$dQ = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy$$

$$d\psi = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy$$

Flow rate through any line joining two points on two streamlines = difference between the respective values of ψ

Physical Significance of Stream Function: Example



For a two dimensional flow field, the stream function ψ is given as $\psi = \frac{3}{2}(y^2 - x^2)$. The magnitude of discharge occurring between the stream lines passing through points (0,3) and (3,4) is **[2013]**

- | | |
|---------|-------|
| (a) 6 | (b) 3 |
| (c) 1.5 | (d) 2 |

Solution: (b)

Flow rate through any line joining two points on two streamlines = difference between the respective values of Ψ

Stream Function (Ψ - psi)

- For 2D, incompressible, and irrotational flow

$$\text{Vorticity} = 2 \times \text{Rotation} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0$$

$$v = -\frac{\partial \psi}{\partial x}$$

$$\frac{\partial}{\partial x} \left(-\frac{\partial \psi}{\partial x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial \psi}{\partial y} \right) = 0$$

$$u = \frac{\partial \psi}{\partial y}$$

$$\nabla^2 \psi = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0$$

(Laplace equation)

Stream Function (Ψ - psi)

- If stream function exists, then it represents a case of possible fluid flow – continuity equation is automatically satisfied
- If stream function exists, then the flow may be rotational or irrotational
- If stream function exists and satisfies the Laplace equation, it is a possible case of irrotational fluid flow
- Streamlines are applicable for both 2D and 3D flows, but stream functions are defined for 2D flow only